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PMC12284613 | CuAAC 点击化学 | 采集状态 imported_unchecked

标题	Double‐Click Strategy Combining CuAAC and (Thia‐) Diels‐Alder Reactions; Application Toward Peptide Labeling
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出处	Chemistry (Weinheim an Der Bergstrasse, Germany) 2025
DOI	10.1002/chem.202501732
原文链接	https://pmc.ncbi.nlm.nih.gov/articles/PMC12284613/
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摘要（原文节选）

AbstractWe report an efficient double‐click strategy combining copper‐catalyzed azide – alkyne cycloaddition (CuAAC) with (thia‐)Diels – Alder (DA) reactions, enabled by newly designed heterobifunctional platforms bearing orthogonal clickable groups: alkyne – dithioester, alkyne – maleimide, or azide – diene. These platforms were evaluated through both one‐pot sequential protocols (CuAAC/DA or CuAAC/thia‐DA) and three‐component reactions (3CR), using model substrates with complementary functionalities. The use of a highly re ⋯

数据表 · Table 1（17 行，含数值行 17 行）

Results obtained in the different one‐pot double‐click approaches[a]

Entry	Double‐click approach	Partners/Sequence	Product	Yield [%]
1	CuAAC/thia‐DA	(S1 + BP1a) + S3	P1a	nd [b]
2	thia‐DA/ CuAAC	(S3 + BP1a) + S1	67	
3	3CR	S1 + S3 + BP1a	nd [b]	
4	thia‐DA/CuAAC	(S3 + BP1b) + S1	P1b	55
5	CuAAC/DA	(S1 + BP2) + S3	P2	39
6	DA/CuAAC	(S3 + BP2) + S1	61	
7	3CR	S1 + S3 + BP2	60	
8	CuAAC/DA	(S2 + BP3) + S4a	P3a	46
9	DA/CuAAC	(S4a + BP3) + S2	30	
10	3CR	S2 + S4a + BP3	49	
11	CuAAC/DA	(S2 + BP3) + S4b	P3b	50
12	DA/CuAAC	(S4b + BP3) + S2	56	
13	3CR	S2 + S4b + BP3	54	
14 [c]	3CR	S2 + BP3 + S5a	P4a	54
15	CuAAC/thia‐DA	(S2 + BP3) + S5b	P4b	65
16	thia‐DA/CuAAC	(S5b + BP3) + S2	63	
17	3CR	S2 + S5b + BP3	43	

脚注（逐字）：[a] Conditions (with 1 equiv BP, 1.2 equiv S). (i) DA or HDA: H2O/iPrOH (1:1), 37 ° C, 1 hour; (ii) CuAAC or 3CR: CuSO4 · H2O (1 equiv), NaAsc (2 equiv), H2O/iPrOH (1:1), 37 ° C, 1 hour. [b] Not determined; the product was not isolated. [c] Modified conditions: CuSO4 · 5H2O (10 mol%), NaAsc (20 mol%), EtO

正文对应句（逐字）：When the reaction sequence was reversed, starting with the thia‐DA followed by the CuAAC reaction, the desired conjugate P1a was successfully obtained as an inseparable mixture of four isomers (in a ratio of 17:14:29:40, measured by 31P NMR spectroscopy) and isolated in a satisfactory yield of 67% (

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